

We used to call analytics
the “*voice of the customer*”

We were interpreting **clicks** as a proxy for what they wanted.

The most valuable **analytics data** in your organisation isn't in your website tags

It's in your AI chatbot logs. And you're probably giving it away.

Two years apart. **Same question.**
Completely different shape of data.



→
TWO YEARS ·
ONE LAYER UP

EDMONBRAIN 2023

Google Chat - Edmonbrain - Chat

Chat Find people, spaces

Edmonbrain

Chat About

Mark Edmondson 1 min
can you summarise the main themes of the Learning Google Analytics book?

Edmonbrain App 1 min
Thinking...

Edmonbrain output

Answer
The main themes of the Learning Google Analytics book are:

- * GA4 is a new website and mobile application measurement and analytics tool that replaces Universal Analytics.
- * GA4 has a new simpler data model that makes it easier to understand and use.
- * GA4 can be integrated with the Google Cloud Platform to open up a whole new world of applications.
- * Although these new cloud services need new skills including coding, the new services of the cloud make this more accessible than in the past due to its development of serverless architecture offering to abstract away a lot of the leg work configuring and scaling up compute services.
- * A general recommendation when starting out is to aim to use services

History is on

Prompts. Answers. Reasoning.
"What were they thinking?"

Same skills. Same data discipline. **Order-of-magnitude richer signal.**

EDMONLYTICA POST

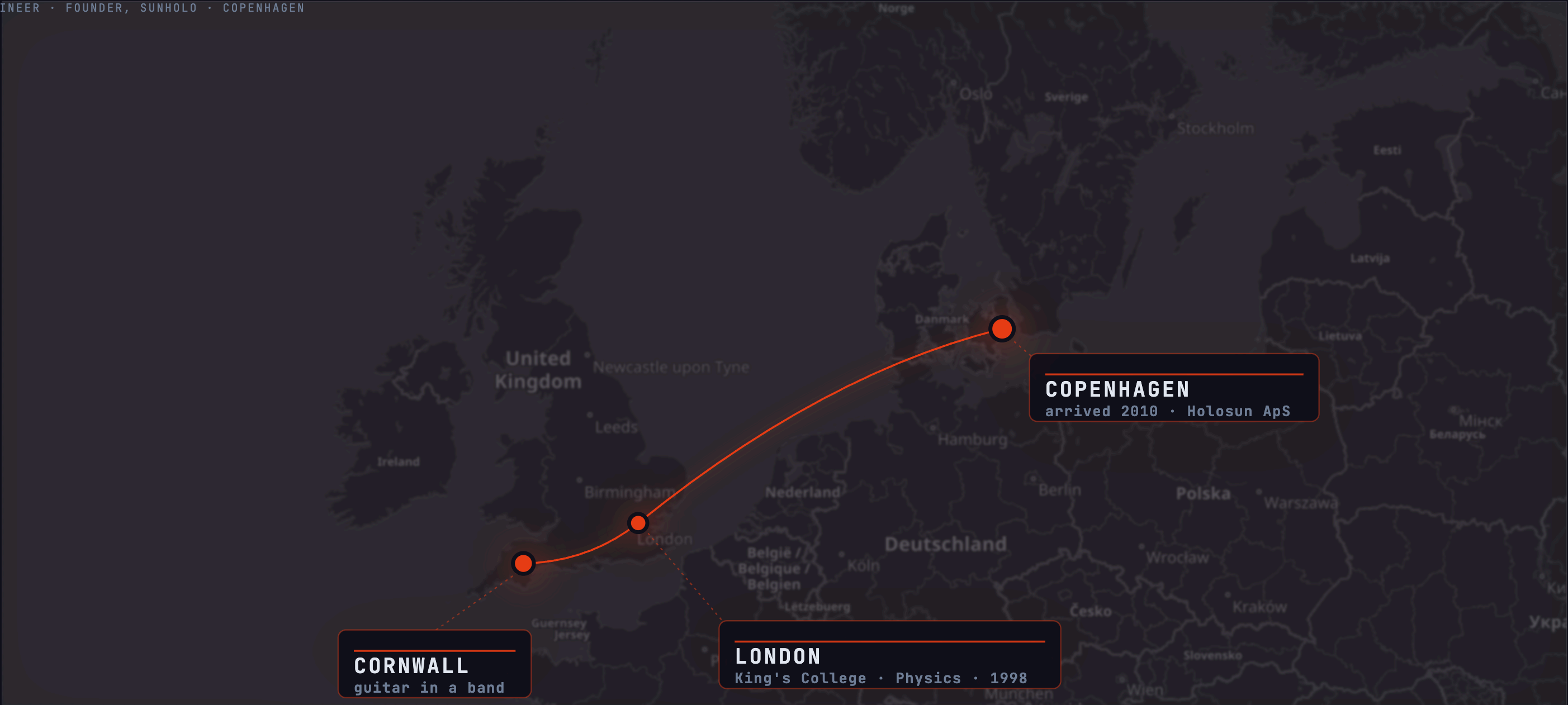
EDMONBRAIN POST

MULTIVAC · THE PLATFORM THAT GREW FROM IT



Mark Edmondson

FORWARD DEPLOYED ENGINEER · FOUNDER, SUNHOLO · COPENHAGEN



THE SAME JOURNEY, ONE LAYER UP EACH TIME



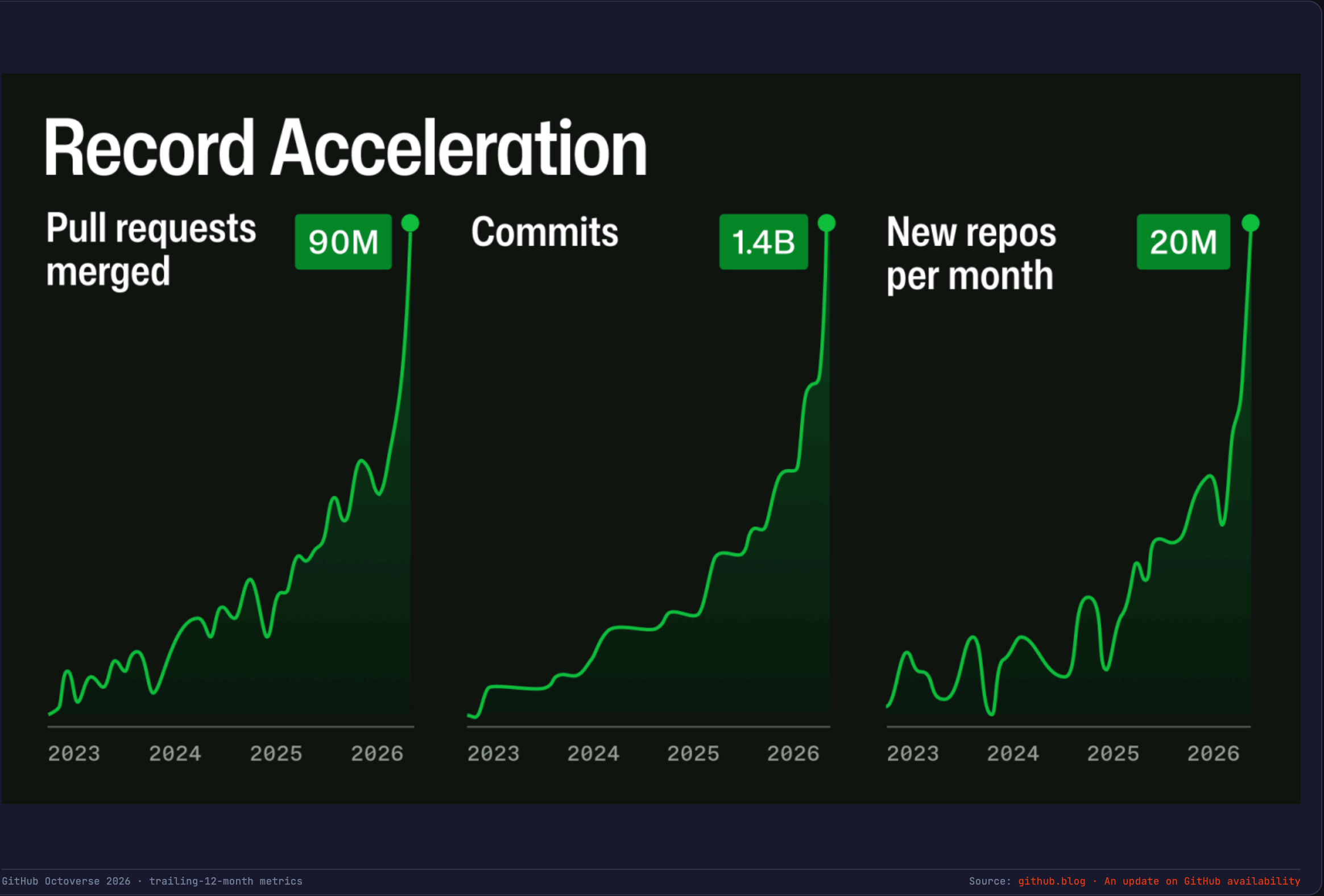
👁 ACT 1 • 6 MIN

What I **Saw** Changing

The paradigm shift, told as personal observation.

I bet this would be **transformational**

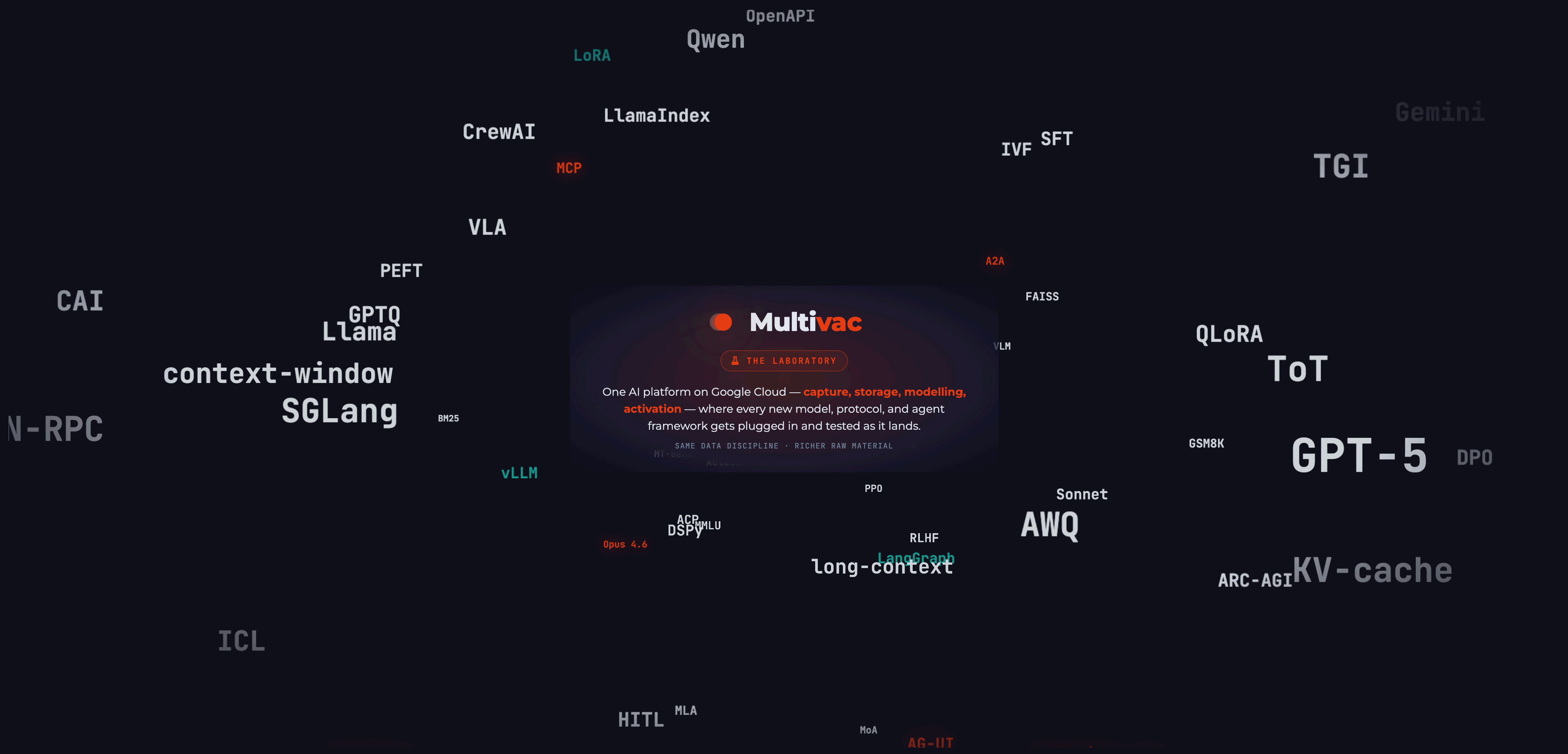
Software output exploding — 90M PRs · 1.4B commits · 20M new repos / month on GitHub.



G Side bet: Google would catch up — full stack already there (TPUs · DeepMind · GCP · BigQuery). Epoch AI · Google controls the most AI computing

1.1 • The Overwhelm

Three years of alphabet soup. Couldn't keep up part-time.



1.2 • The Rosetta Stone

AI is still finding its abstractions. The mature ones already exist — we built them in digital analytics.

STAGE	DIGITAL ANALYTICS	AI / AGENTS
 CAPTURE	events, dataLayer, gtag // what happened?	→ prompts , system message, tool defs // what was asked?
 STORE	warehouse, BigQuery, GA4 export // the lake	→ vector DB , RAG, knowledge graph // pulled into context
 MODEL	attribution, funnels, segments // inferring intent	→ the LLM itself // reading intent
 ACTIVATE	audiences, personalization, bidding // act on it	→ UX, harness, tool calls // agent acts

 Same four stages from *Learning Google Analytics* • O'Reilly 2022 • new raw material, same discipline

1.3 • Same Discipline, Richer Data

Capture the moment, add context, send it somewhere. Look what's new.

GA4 EVENT

gtag.js

```
gtag('event', 'view_pricing', {
  page_path: '/pricing',
  user_id: 'u_8421',
  plan_tier: 'team',
});
```

PAGE

USER

DIMENSION

LANGFUSE TRACE

@langfuse/sdk

```
langfuse.trace({ id: 't_8421', userId: 'u_8421' })
  .generation({
    name: 'pricing_qa',
    model: 'claude-opus-4-6',
    input: 'is this worth it for a small team?',
    output: 'For a 5-person team...',
    usage: { in: 412, out: 186, cost: 0.011 },
    latencyMs: 1820,
  });
```

PROMPT

RESPONSE

MODEL

TOKENS

COST

LATENCY

- Session → Trace
- Pageview → Generation
- Event → Tool Call / Span
- User ID → Thread ID
- Custom dim → Metadata



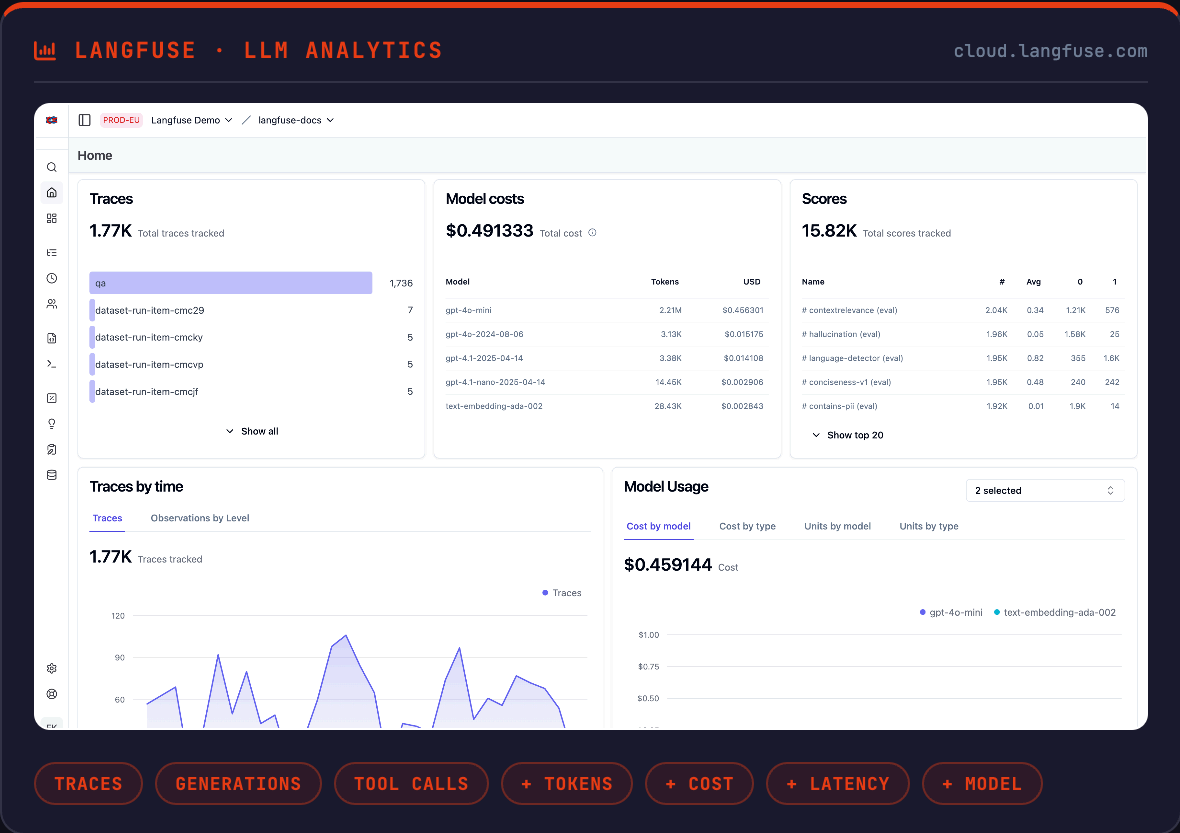
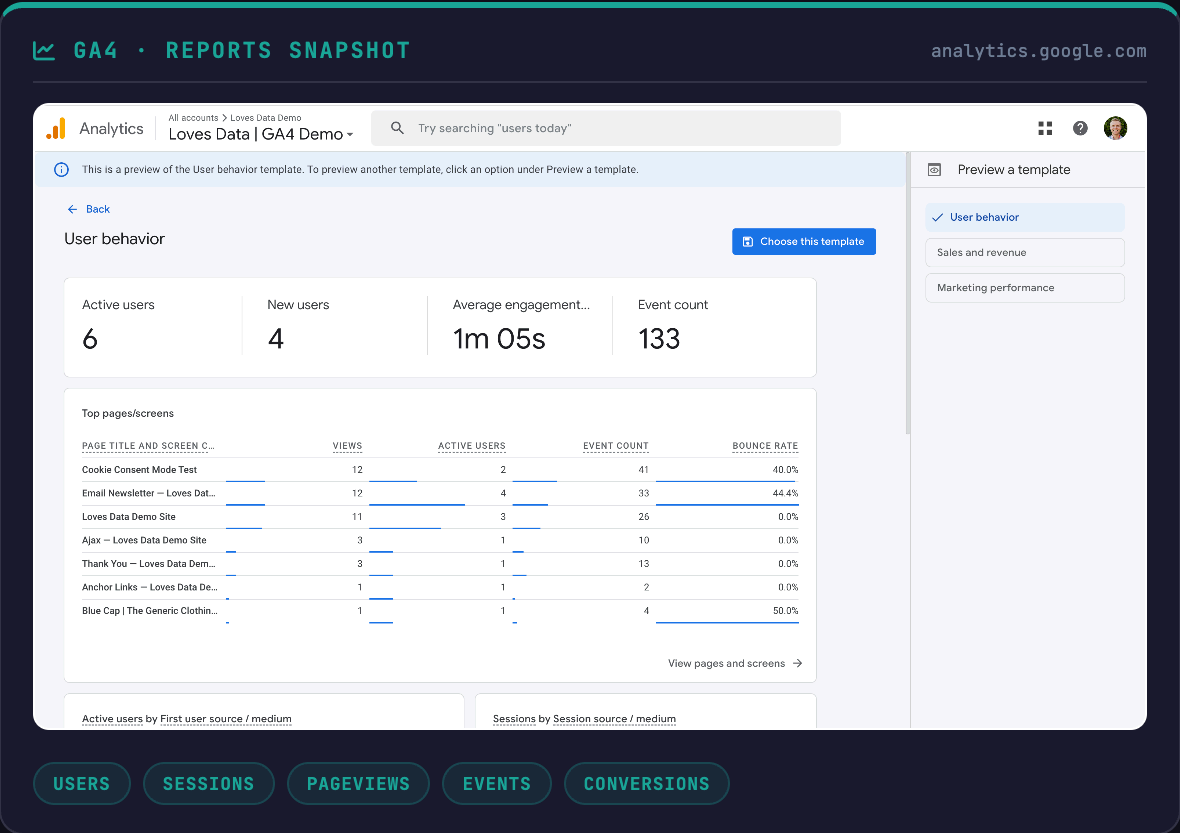
COMING UP TONIGHT • JULIUS FEDOROVICIUS

The Two Pillars of Modern Tagging: Security and Enrichment

"Julius will show you how to harden your tagging tonight. I'm here to ask whether you're capturing the right surface in the first place."

1.4 • The dashboards look familiar too

Same boxes, same charts, same drill-downs — *radically* richer signal inside.

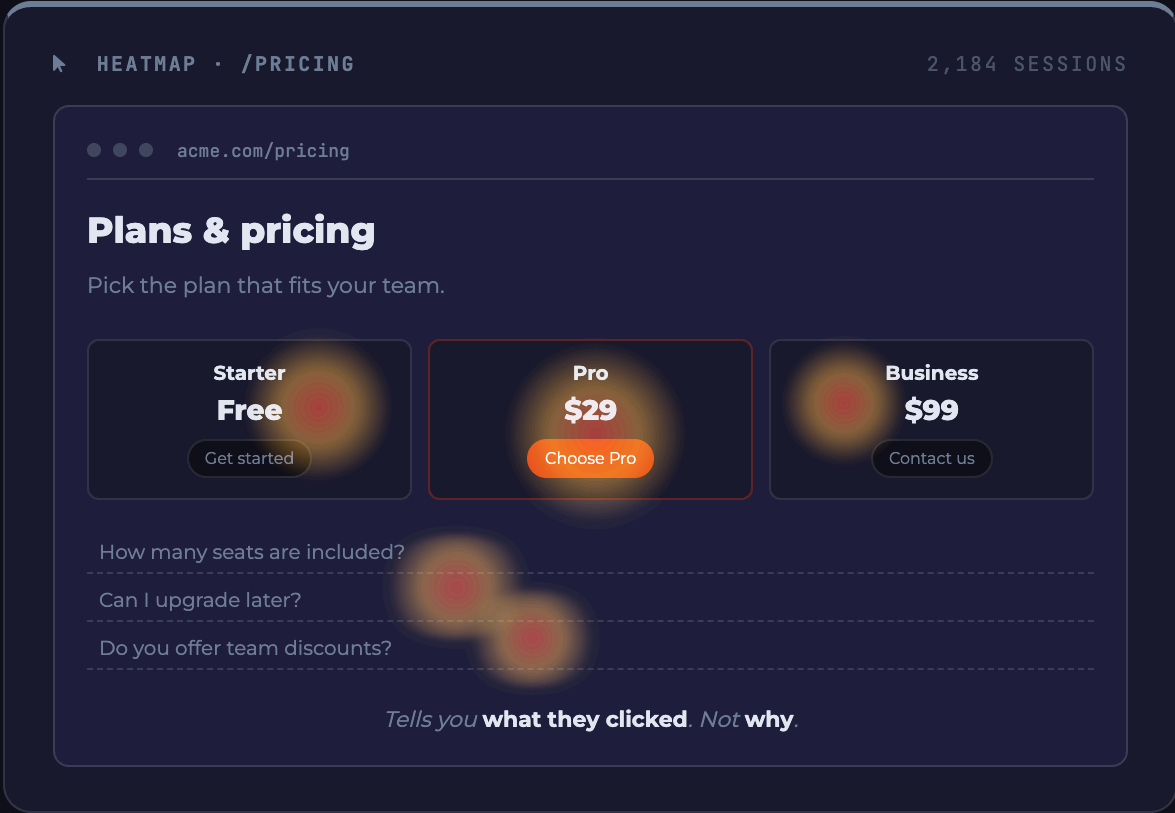


Q ACT 2 · 11 MIN

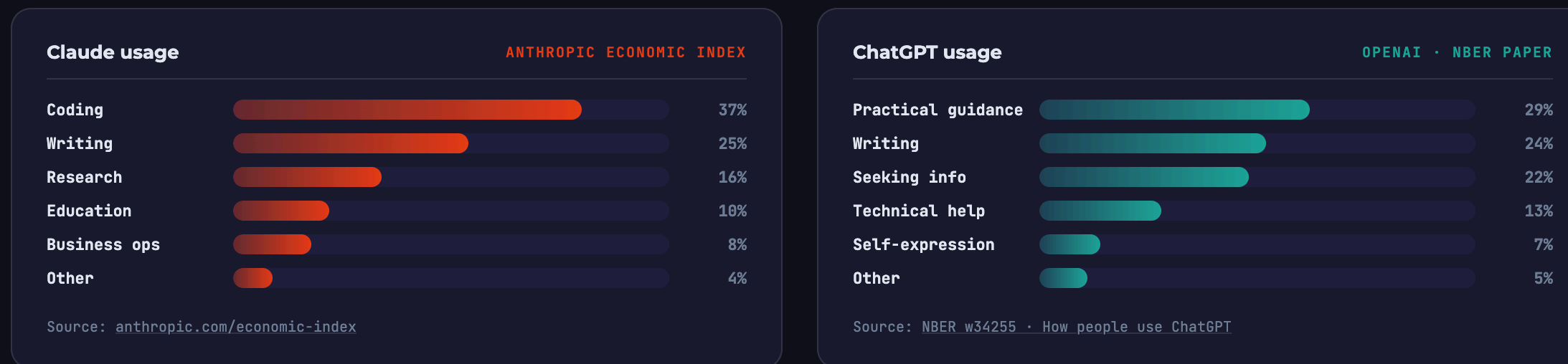
What I Found

The gold mine, told through actual work.

2.1 • Prompts Are Intent

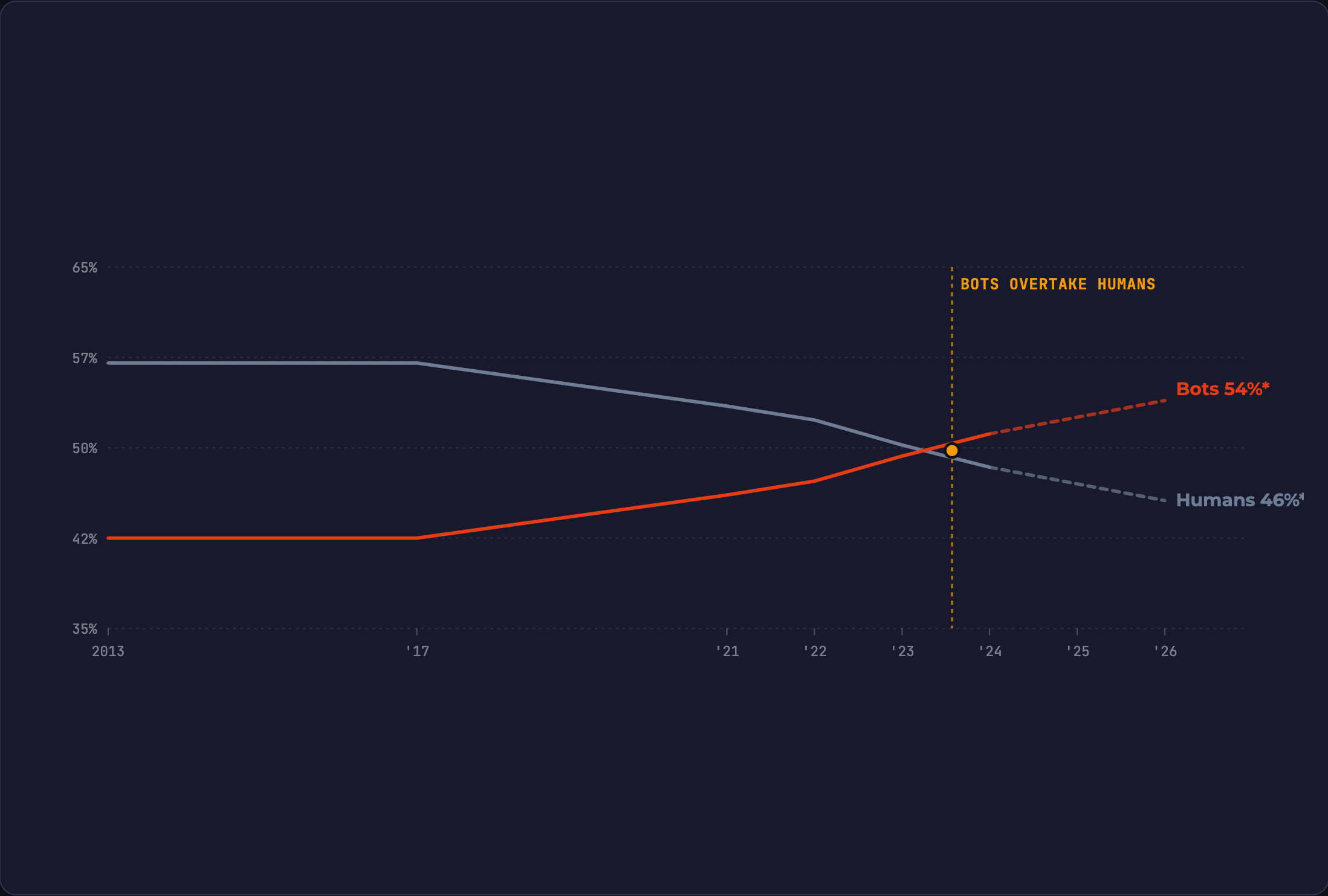


2.2 • **They** Know What Their Users Want



They publish whole roadmaps from this. Do you have this view of your own users?

...and we're at **peak human** internet.



IMPERVA • 2024

51% bots

First time automated traffic surpassed human visitors. Up from 43% in 2013.

🔗 [imperva.com](#) • 2025 Bad Bot Report

IMPERVA • 2013 → 2024

57% → 49%

Human share of web traffic. Declining for the sixth straight year.

🔗 [businesswire](#) • Imperva 13-yr trend

CLOUDFLARE • 2025

+305% YoY

GPTBot crawler growth in one year. 82% of AI crawls are for training, only 2% for user actions.

🔗 [cloudflare](#) • Googlebot to GPTBot

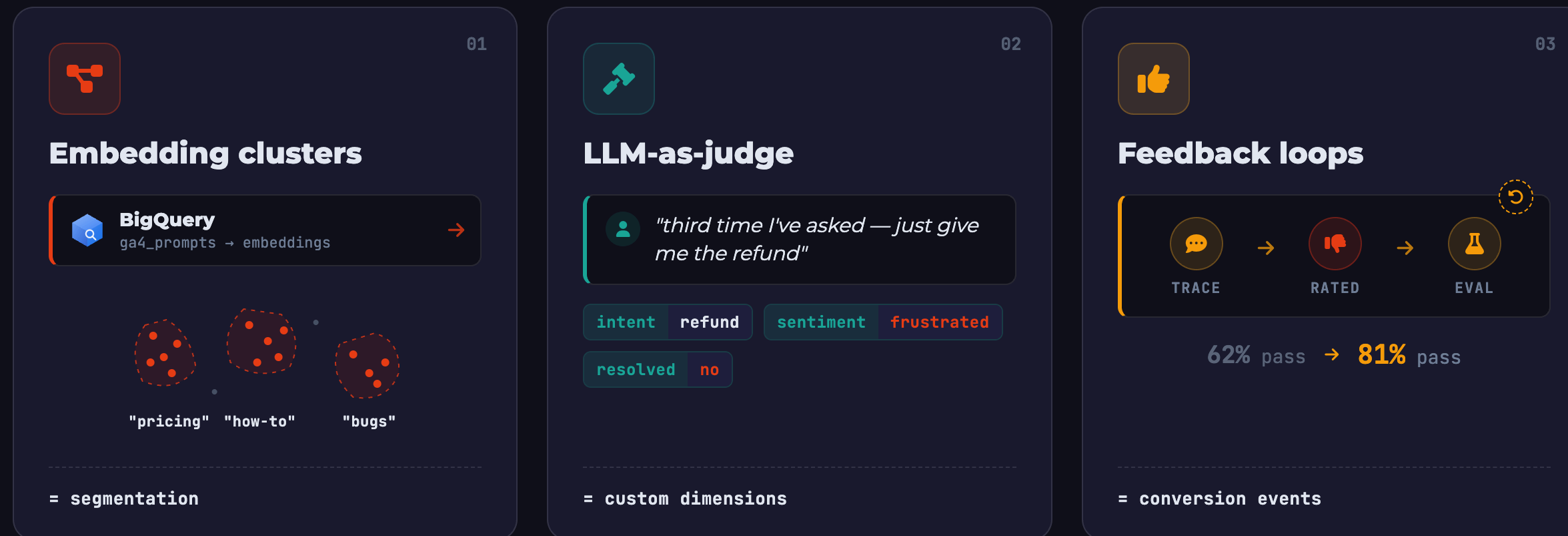
And the prompts? **They're not even on this chart.** They're inside Claude, ChatGPT, Gemini — places your tags can't reach.

2.3 • "We thought everyone was doing X. They were actually doing Y."



2.4 • Analysing **Text at Scale**

How do you actually analyse thousands of prompts? Text, not numbers.



Sound familiar? It's **funnels and segments**. Just different raw material.

2.5 • Langfuse Demo

Same analytics discipline — radically better data.

Langfuse v3.82.0

Langfuse Demo / langfuse-docs / Traces

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

LLM-as-a-Judge

Human Annotation

Datasets

Settings

Support

FK Felix Krauth felix@langfuse.com

Trace: eb346ba8-19d9-4368-9b71-022cd17f25d7

Search (type, title, id)

Timeline

qa

11.58s \$0.000371

retrieval

1.55s \$0.000001

prompt-embedding

0.67s 12 → 0 (\$12) \$0.000001

vector-store

0.69s

context-encoding

0.47s

fetch-prompt-from-langfuse

0.47s

generation

9.57s 1301 → 291 (\$1592) \$0.00037

qa ID

2025-07-16 23:44:16.449

Session: lf.docs.conversation.s7A0W7l

User ID: u-M9QTUzj

Env: default

Latency: 11.58s

Total Cost: \$0.000371

1,313 → 291 (\$1,604)

Release: 044d0162fe3dbddd09f88a554521ddf73726618c

Preview Scores

Formatted JSON

Input

View as markdown

"How to update the input and output in langgraph traces?"

Output

View as markdown

"To update the input and output in Langfuse traces, it's essential to understand that Langfuse allows meticulous tracking of steps within the AI process by logging both inputs and outputs of large language models (LLMs). This means that every interaction you have can be captured and displayed transparently within the UI.

Here's how you can effectively manage and visualize these updates:

Steps to Update Traces

1. **Log Inputs and Outputs**:

Ensure you're logging the inputs and outputs of

...expand (989 more characters)

"

Metadata

pathname: "/docs/demo"

03 • trace

One trace, opened up

retrieval → embed → generate



FROM 2.4 • NOW RUNNING

Judge scores attached



HUMAN SIGNAL

Feedback on the trace



NEW VS GA4

See parallelism & latency shape



COMING UP TONIGHT • PETER MEYER

Using end-to-end testing to verify tracking implementation

"Peter will show you how to verify your GA4 tags actually fire. The same discipline applies to agent traces — except now you're verifying reasoning, not just events."

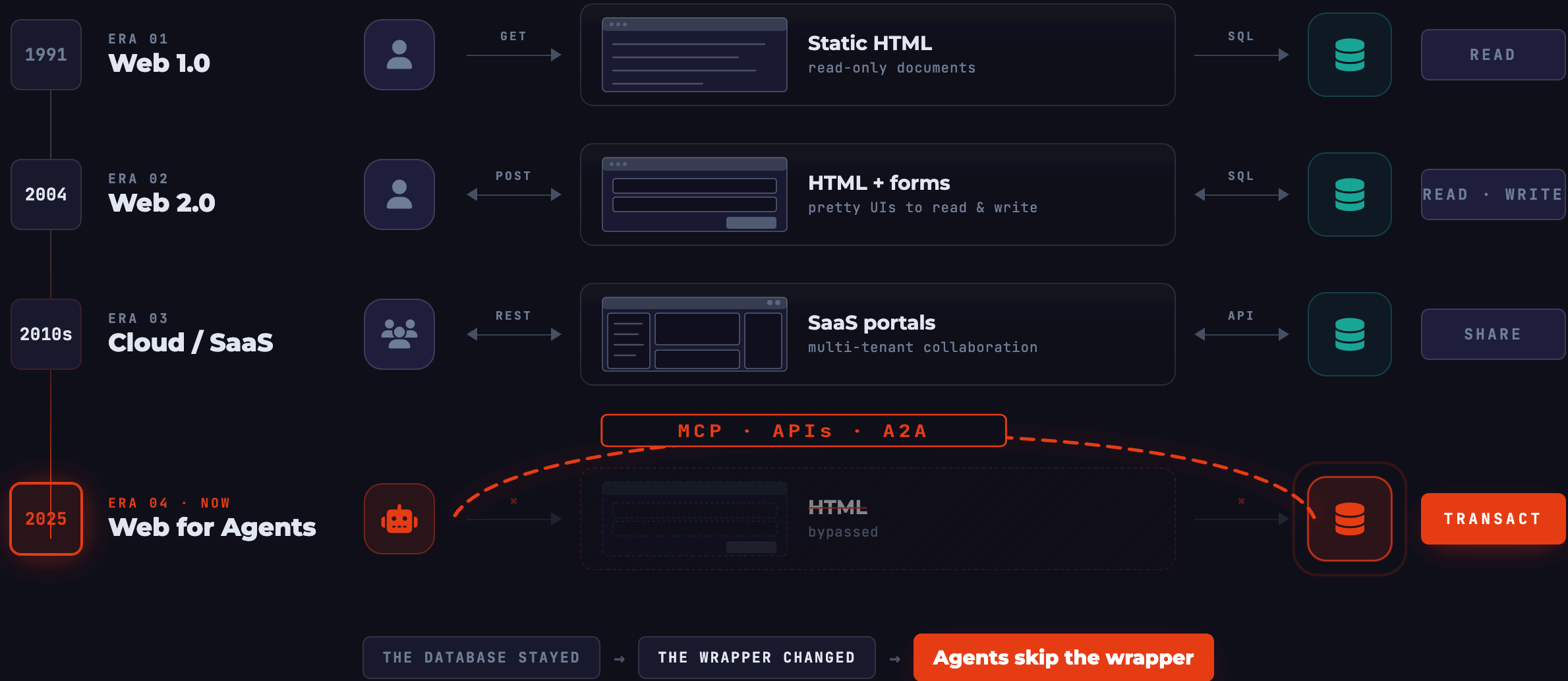
🚀 ACT 3 • 7 MIN

What I Think Is **Coming**

Speculation grounded in experience.

3.1 • The web evolved in eras

What changed isn't the data — it's the path to it.



3.3 • Payment Required is back

HTTP STATUS

402

PAYMENT REQUIRED

Reserved 1991. Unused for 30 years. Revived by AI agents in 2025.

Agent hits a paid endpoint → server returns **402 + price** → agent signs a stablecoin payment → retries with payment receipt header → gets the resource. One round trip. No accounts, no API keys, no human in the loop.

x402

COINBASE • CLOUDFLARE • STRIPE

HTTP 402 revival. Pay-per-request stablecoin micropayments. Cloudflare uses it for pay-per-crawl.

x402.org

AP2

GOOGLE • 60+ PARTNERS

Agent Payments Protocol. Authorization & audit framework. A2A x402 extension is production-ready.

cloud.google.com • AP2

TAP

VISA

Trusted Agent Protocol. Identity & merchant trust signals for agent-initiated card payments.

visa.com • TAP

ACP

STRIPE • OPENAI • MASTERCARD

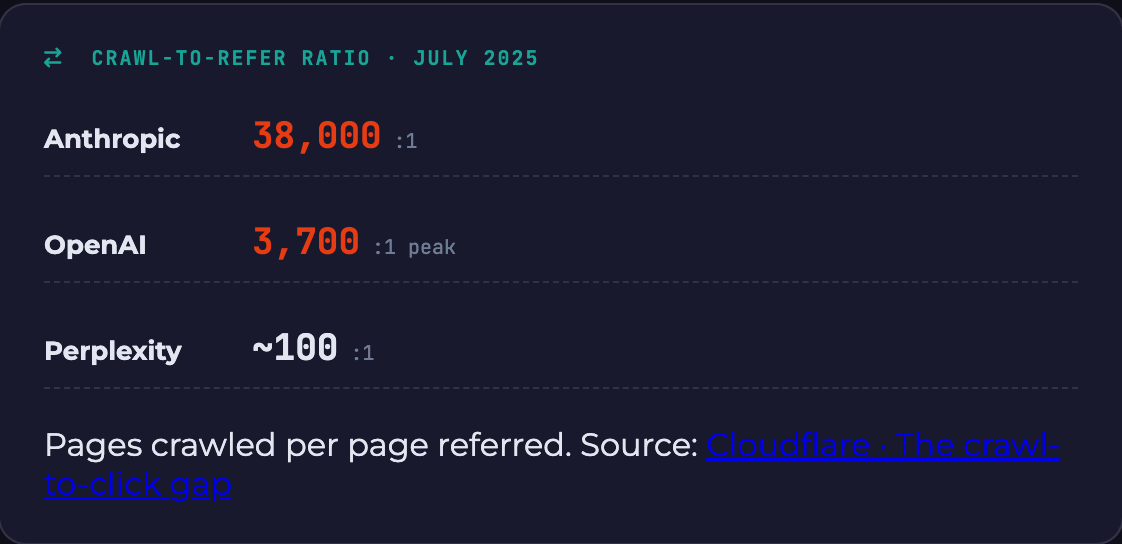
Agentic Commerce Protocol. Open spec for agent checkout in third-party storefronts.

stripe.com • ACP

*Galaxy estimates agentic commerce reaches **\$3-5 trillion** by 2030.*

3.4 • Websites become research surfaces

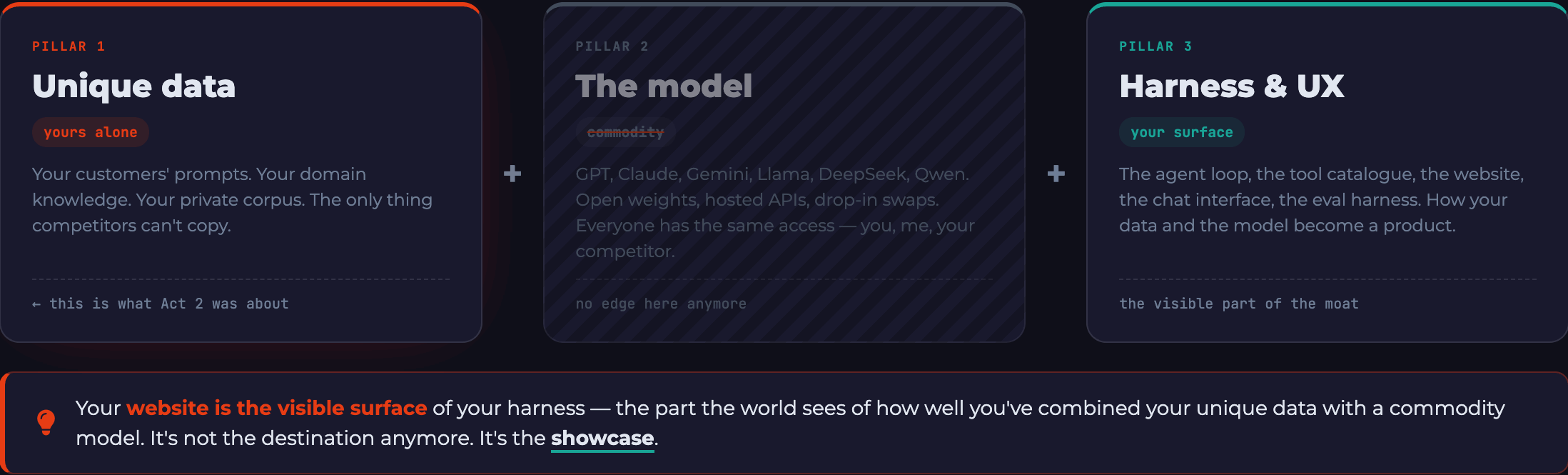
Crawled for what's *not* in the training set. Visited by humans almost not at all.



Your website used to be the destination. It's becoming a **freshness layer** — queried only for what isn't already in the model's head.

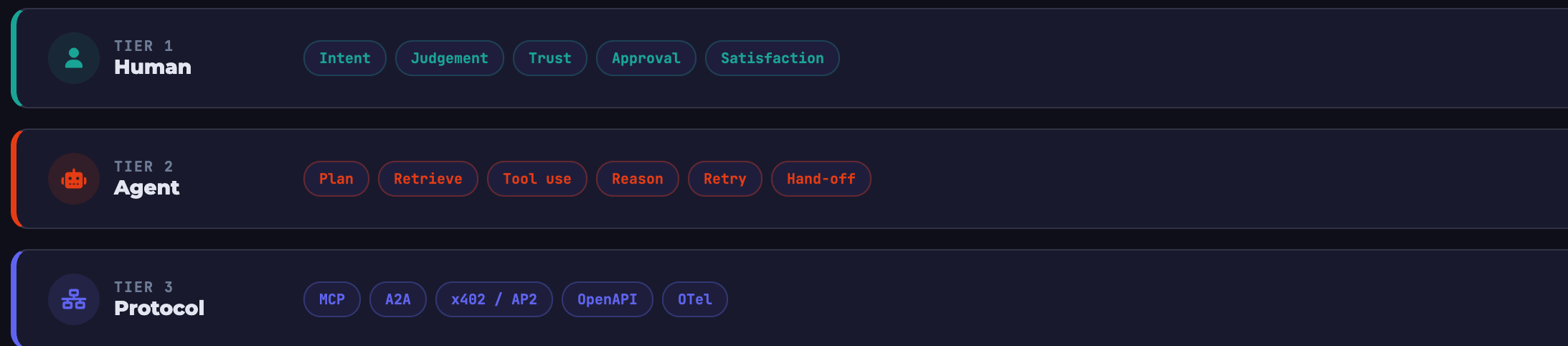
3.5 • The competitive surface

Everyone has the same models now. The moat lives somewhere else.



3.6 • The new **customer journey**

Not a funnel anymore. A stack — with humans on top and protocols at the bottom.



COMING UP TONIGHT • CAROLINE VIDAL • DFDS

Agentic Lead Enrichment & Attribution

"Caroline goes deeper on agentic enrichment after me — she'll show you how to attribute and act on these stratified journeys. I'm here to ask whether you're capturing them at all."

3.7 • A new class of data

Some metrics rename. Most are genuinely new.

WEB ANALYTICS	AGENT ANALYTICS
Pageviews	→ Tool calls
Sessions	→ Traces
Conversion rate	→ Task success rate
Bounce rate	→ Hallucination & retry rate
Cost per acquisition	→ Cost per task (tokens × rate)
Page load time	→ Time-to-first-token / latency budget
Funnels	→ Trace trees
Custom dimensions	→ LLM-as-judge scores
BigQuery	→ Still BigQuery

The tools changed. The thinking didn't.

💡 CLOSE

Three Things I Learned

01

Prompts reveal intent.

I started capturing them.

02

The voice of the customer is louder in prompts than clicks and impressions.

I started analysing it.

03

Embrace the new protocols to take advantage of the new ways AI will shape the web.

I'm evolving with it.

Don't give that **gold** away

“The richest analytics data in your organisation might not be in your website tags anymore. It might be in the conversations your users are having with AI.”

👋 THANK YOU

Let's keep **talking**



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SUBSTACK
markedmondson

Questions, ideas, or want to keep the conversation going? *Any of the above works.*